

Technology Stack

Date	25 June 2026
Team ID	XXXXXX
Project Name	Smart Lender
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	HTML, CSS, Jinja2	Simple, lightweight templates (home.html, predict.html, submit.html) that integrate seamlessly with Flask.
2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Python, Flask	Fast processing and easy integration with the Python-based machine learning data science lifecycle.

3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	CSV files, local OS File System	The loan_prediction.csv handles static training data, while the file system stores the serialized models (.pkl).
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	Local Web Server (Waitress/Gunicorn)	The .pkl files isolate the model from the web server, making local hosting or lightweight cloud hosting highly cost-effective.
5	Version Control & CI/CD <i>(e.g., GitHub, GitLab, Jenkins)</i>	Git, GitHub	Allows the team of 5 members (Nandini, Priyanka, Gopal, Guna, Narayana) to collaborate seamlessly on different scripts.
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	Scikit-learn, XGBoost, imbalanced-learn, Joblib	Enables automated model selection, class balancing (SMOTE), and model serialization for deployment.